Chat Export for sanidhya.rana_contractor@celanese.com

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: I need a material for a circuit breaker bobbin

Assistant: To help identify the best material for your circuit breaker bobbin, could you please clarify:

1. What is the exact function of the bobbin in your circuit breaker? Is it mainly for coil winding support, or does it also serve as part of a relay/crank mechanism?
2. Where will this circuit breaker be used (for example, in residential power distribution, industrial control panels, or another specific environment)?

Once you provide this, I'll ask about wall thickness, preferred base materials, and openness to alternatives to ensure we find the most suitable material.

User: It is a residential circuit breaker.

Assistant: Thank you for clarifying that your circuit breaker bobbin is for a residential application.

Could you please provide the following details:

1. What is the minimum wall thickness of your bobbin part?
2. Do you have any preferred base material for this application (such as a specific polymer type)?
3. Are you open to alternative material suggestions if your preferred material is not

available?

These details will help ensure we identify the most suitable material options for your needs.

User: I need thickness of 0.4mm,I don't have a preference on the material.open to suggestions

Assistant: Here are the key material requirements (CTQs) assumed for your circuit breaker bobbin application:

1. Minimum wall thickness: 0.4 mm (as specified by you)
2. Processing requirement: Injection molding
3. Flame rating: UL94 V-0 at 0.8 mm
4. Glow Wire Flammability Index (GWFI): 960°C (at 0.8–1.6 mm)
5. Comparative Tracking Index (CTI): 400–600 V
6. Relative Thermal Index - Electrical (RTI Elec): ≥130°C
7. Flame retardant feature

Are all of these requirements correct for your application? Please confirm so I can proceed to address additional material requirements.

User: looks good but i also need low warpage and Dimensional stability

Assistant: Thank you for confirming the requirements and adding low warpage and dimensional stability. To ensure we cover all critical aspects, could you please clarify the following:

1. Glow Wire Ignition Temperature (GWIT): Is there a specific value you require, or is standard performance acceptable?

2. Hot Wire Ignition (HWI): Do you need a particular performance level, or is standard acceptable?
3. High-Current Arc Ignition (HAI): Is there a required value, or is standard performance sufficient?

Please let me know your preferences for these, so I can finalize the material requirements and suggest suitable grades.

User: i dont have this information recommend grades based on the req discussed so far

Assistant: Based on your requirements for a circuit breaker bobbin (including thin-wall 0.4 mm, flame retardancy, electrical insulation, low warpage, and dimensional stability), here are potential material options that may be suitable:

1. **CELANEX® XFR 6852 GF30**
   - UL 94 V-0 at 0.4 mm, GWFI 960°C, CTI 600 V, RTI Elec 140°C, 26% glass fiber, halogen-free, excellent dimensional stability and low moisture uptake.
2. **CELANEX® XFR 6842 GF30**
   - UL 94 V-0 at 0.4 mm, GWFI 960°C, CTI 500 V, RTI Elec 140°C, 26% glass fiber, halogen-free, good dimensional stability.
3. **Zytel® FR95G25V0NH NC010**
   - UL 94 V-0 at 0.4 mm, GWFI 960°C, CTI 600 V, RTI Elec 160°C, PA66/6T blend, 25% glass fiber, strong dimensional stability.
4. **Zytel® HTNFR42G30NH NC010**
   - UL 94 V-0 at 0.4 mm, GWFI 960°C, CTI 600 V, RTI Elec 130°C, PPA, 30% glass fiber, high dimensional stability.
5. **Zytel® FR70G30V0NH NC010**
   - UL 94 V-0 at 0.4 mm, GWFI up to 960°C, CTI 600 V, RTI Elec 140°C, PA66, 30% glass

fiber, good dimensional stability.

These grades are selected based on their technical properties and suitability for your application. Please note that final material selection and product design decisions rest with you. These suggestions are based on technical data and do not guarantee suitability for your specific design. Ensure compliance with all applicable standards and regulations. Celanese does not manufacture or certify products for these applications.

If you need colored or sustainability variants, let me know and I can provide options.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.